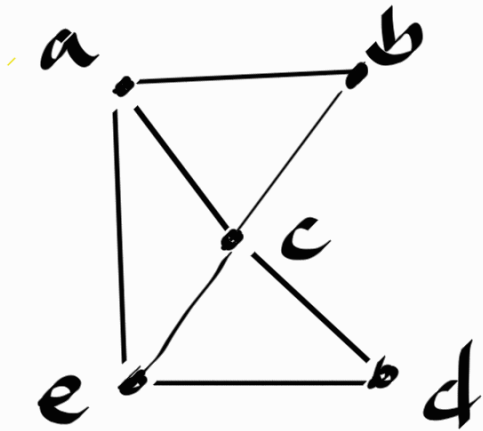
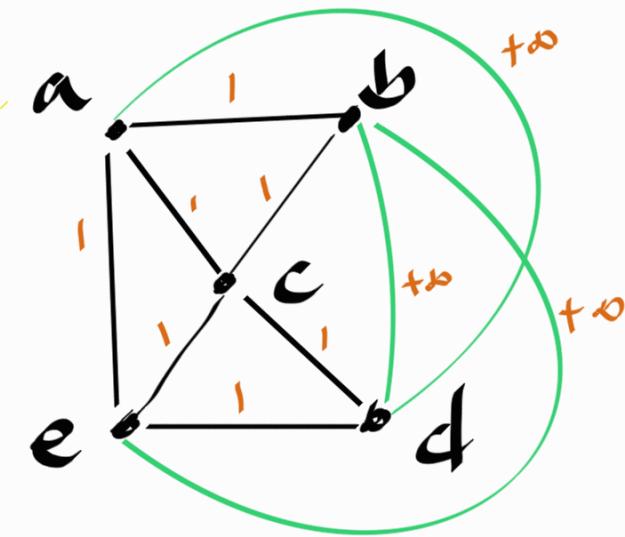


Example:

$HC \leq_p TSP$ (on complete weighted graphs)



$\langle a, b, c, d, e, a \rangle$ HC



Given an input graph $G = (V, E)$, we construct graph $G' = (V', E', w)$, which is a complete weighted graph as follows.

- we keep the original vertices.
- we add new edges between vertices which we did not have edges.
- we give weights to edges as follows.

$$\forall e \in E'$$

$$\text{if } e \in E, w(e) = 1$$

$$\text{if } e \notin E, w(e) = +\infty$$

- This transformation takes $O(n^2)$
 $w \in \text{HC} \iff f(w) \in \text{TSP}$
- Claim! There is a HC in G
iff there is a TSP with weight n in G'

only if $\Rightarrow$

If G has a HC, then use the edges in HC to form a TSP in G' .

As each edge in TSP has weight 1,

The total weight is n .

If part $\Leftarrow$

If G' has a TSP of size n , we know that all the edges with weight ≥ 2 cannot be part of it.

Therefore, all edges in TSP must have weight 1, which in turn means are in C_1 .

Then HC gives us a HC in G .